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Enhanced blind image forgery detection using an accurate deep learning based hybrid DCCAE and ADFC

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Abstract
Currently, digital images are easily manipulated using different image editing tools, and they are difficult to detect. Forgery detection is a hot research topic in security and forensics applications. Several computational methods have been employed to detect blind image forgeries, but efficiency is reduced due to drawbacks such as the lack of appropriate techniques and reduced accuracy. Therefore, this work presents an optimal blind forgery detection using deep learning approaches. Initially, the input forgery images are pre-processed using wiener filtering-contrast limited enhanced histogram equalization (WE-CLAHE) approach. Effective features are extracted from the clustered data using Hybrid dual-tree complex wavelet trigonometric transform (Hybrid DTT) and VGGNet. The feature dimensionality is reduced through improved horse herd optimization (IHH). Finally, a hybrid deep convolutional capsule auto encoder (Hybrid DCCAE) framework is presented for optimal image forgery recognition. The adaptive density-based fuzzy clustering (ADFC) approach is presented to segment similar pixel regions. The accuracy of CASIA V1 dataset is 99.23%, the coverage dataset 98.75% accuracy, and the GRIP dataset is 98.07% accuracy. The deep learning based proposed approach enhances the performance of forgery detection better than the existing approaches.

Keywords Digital images · Blind forgery detection · Deep learning · Image forgery · Adaptive density based fuzzy clustering

1 Introduction

Forgery detection of a digital image is an essential process in digital life since the images are easily manipulated. Presently, digital images are utilized on the internet, print and digital media [37]. The images are mainly considered evidence in legal practices in the

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current scenario. The social media platforms such as Whatsapp, YouTube, Facebook and Instagram are used by people regularly. Any offensive images shared by anyone on the social network are illegal. To check the reliably of shared images in social networks, a methodology tests the genuineness of the image. In a digital world, the possession of visual information becomes cost effective and handy with a fast distribution of inexpensive and simple to utilize devices. Due to the emergence of computers and digital cameras, visual data are easily recorded, stored and frequently shared [13].

Hence, visual manipulation and tampering with visual data is not a complex process to create duplicated images. The illegal image manipulation is known as a forgery [15]. Forgery with a digital image is an uncontrollable issue concerned with the integrity and authentication of the image. The image editing applications can change the images easily; hence the detection and localization of forged images have become a hot research topic [16]. The process of Image authentication is categorized into two approaches. They are the active approach and passive (blind) approach [33]. The active approach is based on the image source, and the blind approach blindly determines an image altering function applied on the image nevertheless to the source of an image. The active approaches are based on information like digital watermarks and signatures [6]. The active approach requires special cameras for inserting the watermark to create digital images. Still, the blind approach can detect the image without needing the original image. Hence it is widely used in forensic applications [25, 35, 39].

Recently, different forgeries are generally applied in digital images like copy-move, splicing, and retouching (enhancing and blurring) [18]. Copy move forgery detection (CMFD) copies a portion of an image and moves the same image to the duplicate portion [43]. Splicing then copies one or more images and moves them to another area of the same image [45]. Retouching is used for increasing or decreasing the image features. Most illegal practices like hiding content, adding information, and growing crowds [5]. Key point and block-based methods are used to recognize and localize the images. Key point approaches are widely used since it extracts only some points in an image. The block-based approach splits the image into more overlapping blocks, leading to time complexity [49].

The processes involved in blind detection are pre-processing, feature extraction, classification (forged or not) and localization. Some of the commonly used image compression techniques to detect forgery are Discrete Wavelet Transform (DWT) [19], principal component analysis (PCA) [42], histogram, geometric moment, DCT, and Singular Value Decomposition (SVD) [30]. Several blind detection approaches have been introduced during the last two decades for identifying the image forgery and determining its reliability in detail. The feature extraction schemes such as Fourier-Mellin Transform (FMT) [48], scale invariant feature transforms (SIFT) [24], geometric moments [26], and DCT [2, 10, 17, 28], etc. have been proposed in the literature.

1.1 Motivation

Nowadays, image forgery has become a major challenge. Digital image forensics is an area that focuses on image authenticity. The forensics verifies the integrity of the image by detecting several forgeries. Digital tamper image detection is necessary globally as the image may get manipulated easily. Forgery images are utilized to defame the people and change the judgment in court. Hence, an effective forgery detection approach is

needed. Various types of forgery are available, like retouching, image splicing, copy and move forgery. Detecting a forged image from the original image is not an easy process. A good forgery detection approach must be effective in manipulations like noise, JPEG compression, scaling and rotation. Traditional techniques like the block-based approach have a computational complexity problem, and the key-based approach cannot extract images for smoothed images.

Further, the machine learning (ML) models take more time to extract features since they utilize conventional methods. As ML approaches are exposed to increased error, requires high storage space, lesser detection accuracy, less convergence rate, high complexity and overfitting issues, better system performance cannot be obtained. The Deep Learning (DL) based models have been widely utilized since the feature extraction is carried out automatically by a neural network and obtain better outcomes when compared to the ML techniques. Hence, in this work, a DL-based blind forgery detection strategy is described to deal with the abovementioned limitations. The major contributions of the proposed works are:

- To detect the forgery image accurately, deep learning based feature extraction approach is considered. As large data set increases the computational complexities, the number of image features needs to be reduced without losing any relevant information. The feature extraction helps to decrease the amount of redundant data. The proposed feature extraction provides valuable features even from the lower contrast areas and the large size images through (Hybrid DTT) and VGGNet due to the efficient learning model.
- A wiener filtering- contrast limited enhanced histogram equalization is presented to preprocess the image effectively. This approach enhances the contrast of images and removes the noise present in the images. The pre-processed input images are fed to the feature extraction process to improve system efficiency.
- A deep learning-based hybrid classification framework called DCCAE is introduced to detect the forgery accurately. This technique is adopted to adversely enhance classification accuracy because of its improved capability.
- Similar pixels are grouped/segmented using the ADFC approach to improve forgery detection performance, and the performances are analysed through various evaluation metrics.

The research article's remaining structure is arranged as follows: section 2 is the recent related research models; Section 3 explains the proposed forgery detection; section 4 discusses the implemented results and, finally, the overall conclusion of work in section 5.

2 Related works

The recent literature works based on blind forgery detection is explained in the following section.

Elsharkawy *et al.* [12] introduced a blind tampering detection method based on holomorphic image processing (HIP) and Support vector machine (SVM). The HIP was used for converting the image to illumination and reflection elements. The illumination element was either constant or varied based on the manipulated or authentic image. Then the hybrid classifiers like SVM and NN were used to classify the image into a normal and tampered image. The performance of the developed model was analyzed on JPEG compressed images and the existence of noise, and the performance was improved than the other approaches.

Vega *et al.* [40] modelled and developed two image forgery detection (IFD) methods. Initially, the spliced images were detected and used ELA (Error Level Analysis) approach highlighted the pixel with several compression stages. At last, the region that does not belong to the original content was pointed out in the image. Then, chromatic interpolation methods were used to detect forgery in a coloured image. The performance was evaluated on the CASIA v11.0 dataset. The outcomes proved that the approach depends on the image quality. The approach for this data set had complexities in identifying the area that didn't belong to the original image.

Siddiqi *et al.* [34] presented a novel IFD approach based on image splicing. The DWT (Discrete Wavelet Transform) with dominant rotated local binary patterns (DRLBP) is used to extract the features. Due to splicing, discontinuities appeared in the image, so DWT divided the image into sub-bands. This model used a robust coding method to decode the DWT sub-bands coefficient. DRLBP was used for determining the local discontinuities. Finally, the SVM was used to detect the forged images.

Barni *et al.* [3] proposed a model for identifying the source and target regions in copy-move problems. At first, a hypothesis testing framework was designed. The aim was to decide the surface between two nearly-duplicate surfaces found by a generic copy-move detector was the real one. Then a multi-branch (convolutional neural network) CNN architecture was designed to solve the hypothesis testing problem using a feature set to exhibit the presence of boundary inconsistencies and interpolation artefacts in the copy-moved region. The implemented framework, trained on a synthetic dataset, achieves better performance on copy-move forgeries.

Jindal [20] proposed CMFD and splicing using Deep CNN and semantic segmentation. Initially, colour illumination was used for applying a colour map once the pre-processing phase was done. The Transfer Learning (TL) method was used to train VGG-16 with two classes from Deep CNN. This model classified the pixel of an image as a forgery or not forgery. Then the classified images were trained by semantic segmentation for localizing forged pixels. The experimentation was carried out on several benchmark datasets and achieved an average accuracy of 98.585.

El-Latif *et al.* [11] proposed a passive model for detecting the image splicing using CNN and DWT. CNN was first used to extract features from spliced images in this work automatically. Then DWT was applied to generate the final features. Then SVM was used to classify the forged images. The experimentation was done on the two benchmark tampered datasets and outperformed other recent published works based on the measures like precision and accuracy. The abovementioned traditional forgery detection approaches are mainly based on hand crafted features for conducting feature extraction and classification. But the discrimination ability and the invariance to the specific transformation of handcrafted features are poor, limiting the forgery detection performance.

Habibi *et al.* [17] presented a blind image splicing detection technique based on color distribution in nearby pixels of edges. The edge pixels were extracted through contourlet transform. Interquartile Range (IQR) is adopted for the precise differentiation of authentic edges from interfered ones which illustrate the histogram distribution of spliced boundaries in YCbCr color space. The proposed segmentation approach in this research adversely improves the localization performance and reduces computational time. Columbia Image Splicing Detection Evaluation (CISED) dataset is used for performance analysis.

Rao *et al.* [29] established a strategy for image forgery detection and localization on the basis of deep convolutional neural network (DCNN). The boundary transition artifacts emerging from blending operations are pervasive in numerous image forgery manage-

ments that are determined by the proposed CRF (conditional random field) based attention framework. The obtained attention maps are adopted to re-weight the convolutional feature maps to minimise noise.

Shah *et al.* [32] introduced a blind copy-move image forgery detection mechanism to check the digital image integrity and genuineness. Initially, a discrete wavelet transform is performed over the input image. The obtained estimated sub-band with low frequency is separated into tiny overlapping blocks holding constant size with one pixel sliding factor. Then 2D discrete cosine transform is evaluated and accumulated as one-row vector through zigzag scanning. To increase the processing speed, both transforms are cojoined with the fast K-means clustering method. Table 1 describes the review of blind image forgery detection techniques with their contribution and merits.

When surveying the existing techniques in forgery detection, there are certain limitations, including high consumption of detection time, complexities in computation, and high availability of noise. Also, there is a requirement for higher storage space due to the larger accumulation of data, and the accuracy is degraded due to the increased occurrence of error. Due to these forms of issues in the surveyed research, accurate detection of blind image forgeries cannot be obtained. Hence, to overcome these limitations and enhance the classification accuracy, a DL model based clustering approach is used to tackle these problems in the proposed research.

3 Proposed methodology

This work explains the overall working procedure of the proposed forgery detection. In this work, copy move and splicing forgeries are identified. Several editing devices can do image forgery. Hence, this work aims to determine the forgery part present in an image. Three datasets are used to verify the effectiveness of the system. Figure 1 shows the framework of the forgery detection. The initial stage for selecting and localizing image forgery is pre-processing [46]. It is used for improving the classification performance in image forgery. Then feature extraction and selection processes are involved. Then the dataset is split into training (70% images) and testing (30% images). Then, the images are trained and tested. Finally, the segmentation is carried out to find the forgery region. The major advantage of the proposed Hybrid DCCAE is that this model can identify multiple forgery images.

The proposed framework is highly novel as it uses an effective feature extraction method for extracting the most discriminant features to minimize the error possibilities. The classification and segmentation techniques are highly novel as a hybrid methodology is adopted to promote better outcomes in blind forgery detection. As the existing methodologies suffer from increased detection time, degraded accuracy and requirement of high storage space, expected outcomes cannot be attained, which can be fulfilled through the proposed work by efficient forgery detection. The description of each section in forgery detection is represented in detail as follows.

Pre-processing by wiener filter-CLAHE The input image size of CASIA V1 dataset is 256×384×3, size of the coverage dataset is 519×294, and 768×1024×3 is the input image size for Grip dataset. The main objective of image pre-processing is to enhance image quality and make a better analysis. The unwanted distortions can be suppressed, and image

Table 1 Analysis of blind image forgery detection techniques

Author name and reference	Technique used	Contribution	Merits	Demerits	Performance
Elsharkawy *et al.* [12]	SVM-NN	To effectively classify normal, tampered images based on HIP and hybrid classifiers.	Noise is reduced to a greater extent and improves system performance.	Less generalization capability.	Accuracy-96.93%
Vega *et al.* [40]	ELA and chromatic interpolation	To highlight the pixel with numerous compression stages and detect a forgery in a coloured image.	High speed and reduced cost of time.	Area identification is complex in the case of images other than the original image.	Accuracy-73.3%
Siddiqi *et al.* [34]	DWT with DRLBP	To present IFD model based on image splicing for detecting the forged images.	High robustness can be obtained, and local discontinuities can be effectively determined.	Accurate detection cannot be obtained.	FPR-2.05, AUC-99%
Barni *et al.* [3]	Multi-branch CNN	To determine the source and target regions in copy-move problems and decide the surface among two nearly-duplicate surfaces.	Better performance on copy-move forgeries can be obtained.	Localization cannot be done effectively.	MSE accuracy- 59.84 for the Grip dataset
Jindal [20]	CMFD and Deep CNN	To undergo slicing and semantic segmentation for improved forgery detection.	Better localization of forged pixels can be seen.	Increased time complexities.	Accuracy-98.58
El-Latif *et al.* [11]	CNN and DWT	To detect image slicing with improved performance.	The system capability is fine.	Poor discrimination ability and specific transformation invariance of handcrafted features.	Accuracy- 90.01% for CASIA v1.0 and Accuracy- 96.36% for CASIA v2.0
Habibi *et al.* [17]	Colour Edge Inconsistency using Statistical Dispersion Measures.	To present a blind image splicing approach for efficient performance.	Localization performance is improved and is highly effective.	Reduced time of computation.	Accuracy- 97.08

Table 1 (continued)

Author name and reference	Technique used	Contribution	Merits	Demerits	Performance
Rao *et al.* [29]	CRF based attention model	To establish effective image forgery detection and localization.	Adverse noise reduction is attained.	Requires high storage space due to larger accumulation of data.	Robustness= 74% when Q factor=75
Shah *et al.* [32]	DWT and 2D discrete cosine transform	To check the digital image integrity through blind copy-move image forgery detection mechanism.	The speed of processing is high.	Degraded classification accuracy due to high chances of error occurrence.	Recall-98.18%

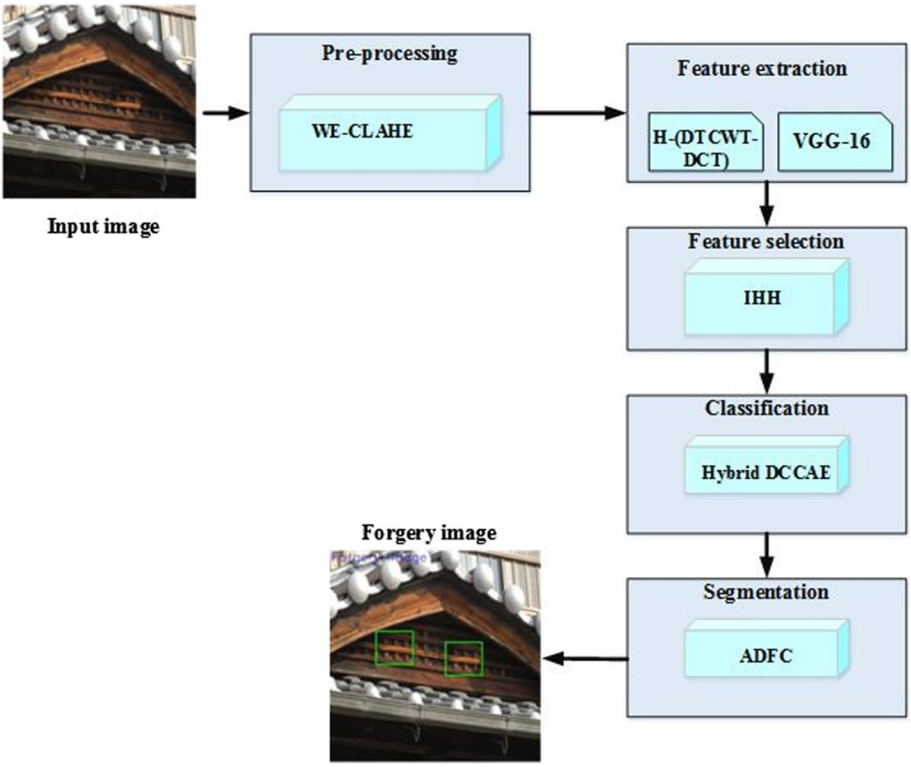


Fig. 1 Framework of the proposed forgery detection

features can be improved to enhance the forgery detection accuracy. Initially, pre-processing is carried out to remove noise and improve the contrast of the images. The wiener filter-CLAHE do the pre-processing process. Wiener filter is a low pass filter, and it works on a statistical model which considers that noises and signals are stationary. It computes the variance and means of the neighbourhood, and if the variation is maximum, it applies minimal smoothing, and if the variation is minimum, it applies strong smoothing. This filter reduces the error between the original and the estimated signal. Let us consider the original image as M and the estimated image as $M^{'}$, then the error measure is,

$$E^2 = e\left\{ \left(M - M'\right)^2 \right\} \tag{1}$$

where e is an expected term. Then the minimum error function is expressed as:

$$M'\left(f_1,f_2\right) = \left[\frac{B^*\left(f_1,f_2\right)S'\left(f_1,f_2\right)}{\left|S'\left(f_1,f_2\right)\right|^2 B\left(f_1,f_2\right) + S'\left(f_1,f_2\right)} \right] C\left(f_1,f_2\right) \tag{2}$$

where $M^{'}(f_1f_2)$ is an estimated image in a frequency, $B(f_1,f_2)$ is a degraded function transform and $C(f_1,f_2)$ is a degraded image transform. Then the term $B^{*}(f_1,f_2)$ is a conjugate of $B(f_1,f_2)$ and $S^{'}(f_1,f_2)$ is a power spectral value of a non-grading image. Once the noise is

removed, CLAHE further improves the contrast of the input. To achieve the image contrast, CLAHE splits the image into tiny overlapping tiles $P \times Q$. A histogram is then performed for each tile. For computing the histogram, initially, the average number of pixels per region is computed by the following expression:

$$Q_A = \frac{Q_x \times Q_y}{Q_g} \tag{3}$$

where Q_x and Q_y are the number of pixels in the x and y dimensions and Q_g is a grey level. Further, the clipping limit is used for clipping the histogram, and it is given by:

$$Q_{cl} = Q_A \times Q_{ncl} \tag{4}$$

where Q_{cl} is clipping limit and Q_{ncl} is a normalized clipping limit. Then, for every tile, the clipping limit is performed for histogram height, and it is given as:

$$H_t = \begin{cases} Q_{ncl} & \text{when } Q_i \geq Q_{ncl} \\ Q_i & \text{else} \end{cases} \tag{5}$$

where H_t is a histogram height of tth tile and the entire clipped pixel is given by

$$Q_c = \left(Q_x \times Q_y\right) - H_t \tag{6}$$

where Q_c is an amount of clipped pixel. Then, the pixels are redistributed, and finally, a cumulative histogram is given by:

$$C_t = \frac{1}{Q_x \times Q_y} H_t \tag{7}$$

Finally, the Rayleigh distribution provides the quality and brightness of an image, and it is given as:

$$g = g_{\min} + \left[2\left(\gamma^2\right)\ln\left(\frac{1}{1 - p(F)}\right)\right] \tag{8}$$

where γ is a Rayleigh variable, $p(F)$ is a CDF (Cumulative Distributive Function) $g_{\min}$ is a minimum pixel value.

Feature extraction using H-(DTCWT–DCT) and VGG Feature extraction is the process carried out to reduce the dimensionality where a huge number of pixels are described effectively so that significant image parts can be gathered. The DTCWT is a recent enhancement to discrete wavelet transform with significant additional properties. VGG-16 is the advanced form of convolutional neural network with higher learning capability and easy implementation. The most significant features can be extracted from VGG 16, possessing 16 layers of deep neural networks. The feature extraction process includes abstraction of significant features from DTCWT and VGG-16. The image quality is enhanced through pre-processing technique, whereas the low contrast areas are improved. Then, the valuables features are effectively extracted from low contrast areas by utilising an effective feature extraction technique called H-(DTCWT–DCT) and VGG-16, reducing the feature dimensionality for improved performance. DTCWT overcomes a shift invariance problem and directional selectivity in 2 or more dimensions. It achieves directional selectivity with the help of analytic wavelets. This model can generate six directional sub bands oriented in the

directions like $\pm 15, \pm 45, \pm 75$ real and imaginary parts. Let $h_j(m)$ and $g_j(m)$ are the filters. The jth phase of the first $F_{new}^{(l)}\left(e^{kw}\right)$ and second $S_{new}^{(l)}\left(e^{kw}\right)$ filter bank response is given as

$$F_{new}^{(l)}\left(e^{kw}\right) = F\left\{FI_{new}^{(l)}\right\}\left(e^{kw}\right)\Big\} \tag{9}$$

The DTCWT is used to decompose the image $g(a,b)$ for the series of translations and dilations of complex scaling and 6 CWT functions are given as:

$$g(a,b) = \sum_{m\in Z^2} a_{jo}, m^{\theta jo}, m^{(a,b)} + \sum_{m\in Z^2}\sum_{k\geq ko}\sum_{m\in Z^2} a_j^{\theta}, m^{\theta_j^{\varphi}}, m^{(a,b)} \tag{10}$$

where a_{jo} and a_j^{θ} are the coefficients of scaling and wavelets. $m^{\theta jo}$, $m^{(a,b)}$ is scaling term and $m^{\theta_j^{\varphi}}, m^{(a,b)}$ is a 6 wavelet term. Finally, the image $g(a,b)$ divided into s non overlapping sub-bands. Then, the centre and highest frequency sub-bands are split into 4x4 blocks. Then the transform function DCT is applied to every sub-bands block. DCT transform is expressed as:

$$g(c,d) = \sqrt{\frac{2}{M}}\sqrt{\frac{2}{N}}\alpha_c\alpha_d \sum_{c=0}^{M-1}\sum_{d=0}^{N-1} I(a,b)\cos\frac{(2a+1)c\prod}{2M}\cos\frac{(2b+1)d\prod}{2N} \tag{11}$$

Finally, H-(DTCWT–DCT) is used to extract the coefficients of H-(DTCWT–DCT) from the pre-processed images.

Feature extraction by VGG-16 It is said to be an excellent vision network, and it has 16 layers. The input size of VGG is 3x224x224 pixel, and it comprises 41 layers (13 convolutional layers, 3 FC (fully connected layers), and 16 layers with learnable weights). It comprises convolutional 3x3 filters with stride 1 and max-pooling layers of a 2x2 filter of stride 2. The first and second convolutional layer has 64 and 128 filters. Other convolutional layers have 256, 512 and 515 filters. Three FC layers are provided at the end, and the first and second FC has 4096 neurons. The final FC is used for compressing features to a thousand dimensions. It is a huge network, and it comprises of 138 million parameters approximately. The architecture of VGG-16 is described in Fig. 2.

H-(DTCWT–DCT) has a characteristic of better directional selectivity, which decomposes the texture image from several directions and achieves multiple feature information descriptions. Further, in VGG16, convolution and max-pooling are applied for the first 4 blocks. Hence, for the input image of size 3x224x224, the obtained feature (output) will have the size of 16x16x512.

Feature selection using IHH After extracting the features, their dimensionality is reduced using IHH. Feature selection is an important process responsible for the optimal gathering of features to improve detection accuracy, minimise the error, and reduce the feature dimensionality that increases the detection speed. Metaheuristic algorithms can extract the essential features from a set of features. This optimization is based on the characteristics of horses, and it has the stages like Grazing (G), Imitation (I), Roam (R), defence (D) and Sociability (S). The movement is given to the horse at every iteration based on the following expression:

$$Y_n^{iter,age} = \overrightarrow{W}_n^{iter,age} + Y_n^{(iter-1),age} \tag{12}$$

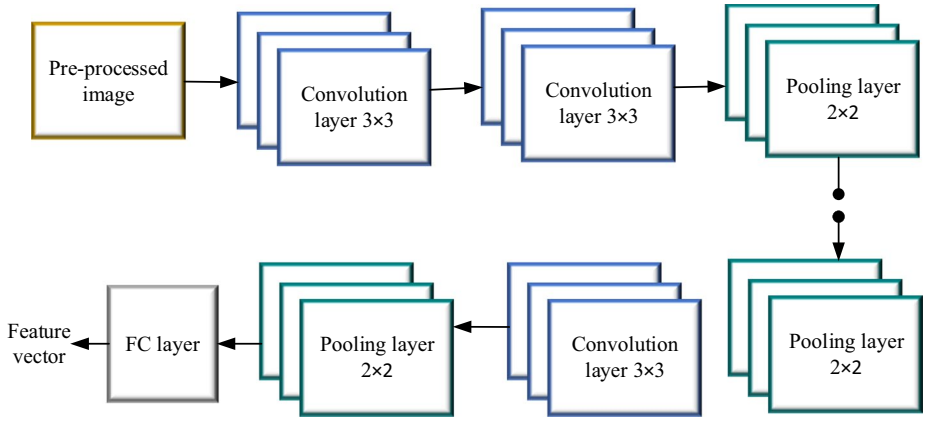


Fig. 2 VGG-16 architecture

where $Y_n^{iter,age}$ is a nth horse position, *age* is horse's age, *iter* is a present iteration and $\overrightarrow{W}_n^{iter,age}$ is a horse velocity vector.

Grazing (G): IHH optimization creates the grazing field around every horse with a coefficient h , and the mathematical model of grazing is given by:

$$\overrightarrow{Gr}_n^{iter,age} = h_{iter}(LL + UL\rho) + Y_n^{(iter-1)} \tag{13}$$

$$h_n^{iter,age} = h_n^{(iter-1),age} \times \sigma_g \tag{14}$$

Where $\overrightarrow{Gr}_n^{iter,age}$ is the motion variable of a horse, and this term is used for reducing the linearity with σ_g iterations. *LL* and *UL* are the lower and upper limit of space in grazing.

Hierarchy (H): Research has proved that the horses follow the hierarchy law in middle age β_i and γ_i. It is expressed as:

$$\overrightarrow{N}_m^{iter,age} = n_m^{(iter-1),age}\left[Y_*^{(iter-1)} - Y_n^{(iter-1)}\right] \tag{15}$$

$$n_m^{iter,age} = h_n^{(iter-1),age} \times \sigma_n \tag{16}$$

where $\overrightarrow{N}_m^{iter,age}$ is an effect of the better location of the horse and $Y_*^{(iter-1)}$ is a better location of the horse.

Sociability (S): It is noted that the horses between the age of 5 to 15 years show interest in a herd, and it is represented as:

$$\vec{P}_n^{iter,age} = p_m^{\left(iter,age\right.}\left[\left(\frac{1}{M}\sum_{k=1}^{M} Y_k^{(iter-1)}\right) - Y_n^{(iter-1)}\right] \tag{17}$$

$$p_m^{iter,age} = p_m^{(iter-1),age} \times \sigma_p \tag{18}$$

where $\vec{P}_n^{iter,age}$ is a motion of social vector and $p_m^{iter,age}$ is an orientation of horse. $p_m^{iter,age}$ minimizes in every cycle with the factor σ_p.

Imitation (I): This characteristic of a horse is regarded as a term j in an algorithm. Small horses portray other horses, and this behaviour doesn't change throughout their lives.

$$\vec{I}_n^{iter,age} = I_m^{\left(iter,age\right.}\left[\left(\frac{1}{Q_n}\sum_{k=1}^{Q_n}\vec{Y}_k^{(iter-1)}\right) - Y^{(iter-1)}\right] \tag{19}$$

$$I_n^{iter,age} = I_n^{(iter-1),age} \times \sigma_i \tag{20}$$

where $\vec{I}_n^{iter,age}$ is a motion vector with $\vec{Y}$ position. Q_n is a total horse with the best position and σ_i is a reduced factor per cycle.

Defence: The defence system in IHH is defined using the negative coefficients in Eqs. (21), (22) used to move the horse away from an unsuitable location.

$$\overrightarrow{D}_n^{iter,age} = -d_n^{(iter,age)}\left[\left(\frac{1}{Q_n}\sum_{k=1}^{Q_n}\vec{Y}_k^{(iter-1)}\right) - Y^{(iter-1)}\right] \tag{21}$$

$$d_m^{iter,age} = d_n^{(iter-1),age} \times \sigma_d \tag{22}$$

where $\overrightarrow{D}_n^{iter,age}$ is an escape vector and Q_n is a horse with the worst positions.

Roam (R): Horses are used for grazing and roaming to search for food from one place to another. Roaming behaviour is generally seen in young horses, and it is shown in the following equation:

$$\vec{R}_n^{iter,age} = r_n^{(iter,age)}\rho Y^{(iter-1)} \tag{23}$$

$$r_m^{iter,age} = r_n^{(iter-1),age} \times \sigma_r \tag{24}$$

where $\vec{R}_n^{iter,age}$ is a velocity vector for a local search and σ_r is a reduced term. In IHH, the following expression represents the velocity of horses between the ages 0 to 5 years.

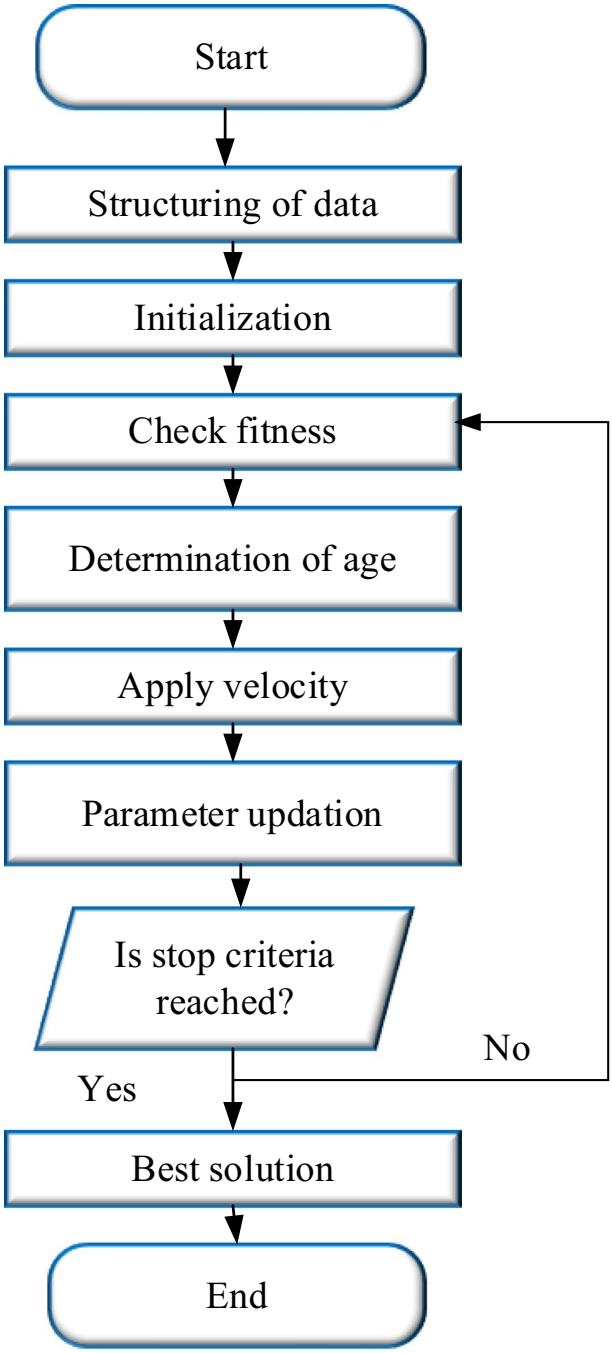
$$\begin{aligned} V_m^{iter} = \beta_i \times \left[g_m^{(iter-1),\delta_{wg}}(u+\rho l)\left[X_m^{(iter-1)}\right]\right) + (w-1) \\ \times \gamma_i\left[i_m^{(iter-1),\delta_{wi}}\right]\left[\left(\frac{1}{pN}\sum_{j=1}^{pN}X_j^{(iter-1)}\right) - X^{(iter-1)}\right] + \\ \left[r_m^{(iter-1,\delta_{ww}}\rho X_m^{(iter-1)}\right] \end{aligned} \tag{25}$$

The regulating parameters β_i and γ_i are utilized for selecting the meaningful features. Hence, optimal characteristics are selected by IHH. The flowchart for IHH is described in Fig. 3.

A detailed description of the feature selection method is given in a flow chart. After updating the parameter, if the evaluation criteria are satisfied for the proposed performance, the optimal solution can be gathered or else the process is repeated until the best criterion is satisfied. The optimal features improve the overall blind forgery detection performance and enhance detection speed.

Classification by Hybrid DCCAE CNN is a multilayer neural network used for 2D data like images. Several regions of the image have been considered input for hierarchical neural

Fig. 3 Flowchart for IHH



network layers, and every layer is extracted using digital filters. Auto encoder is an approach of unsupervised learning technique for neural networks to learn an effective representation of data through network training to eradicate unnecessary distortions. Capsule Auto encoder is combined to adversely enhance the detection accuracy. The proposed architecture can solve the problems of consumption time, lesser rate, and high feature dimensionality. The chances of error occurrence are low, and it improves the detection performance of blind image forgeries. Figure 4 shows the architecture of the proposed Hybrid DCCAE. CNN comprises three layers: Convolutional, pooling, and FC (Fully connected). CNN structure has a different convolution layer that convolutes the input image and the filter.

Convolutional layer An initial layer obtains inputs and extracts the features. The single output matrix is computed by,

$$B_k = f\left(\sum_{j=1}^{M} J_j \times L_{j,k} + C_k\right) \tag{26}$$

where J is an input vector, L is the respective convolution kernel having the size $C_k \times n$. Then, all the convoluted matrices are summed, and C_k bias is added.

Pooling layer This layer is used to minimize the size of the feature maps. Further, this layer has generalization capability, and there are some kinds of pooling operations. This operation split an image into a group of non-overlapping rectangles. The rectangle's largest number is used to produce the output feature map in the max-pooling layer. Likewise, when average pooling is performed, the mean values of the pixel are used to construct a feature map. For input, $y_i \in R^m$ the average pooling $x_i \in R^n$ as,

$$z_i = \frac{1}{r}\sum_{k=0}^{r-1} y_{ri} + k \tag{27}$$

where r is the pooling support

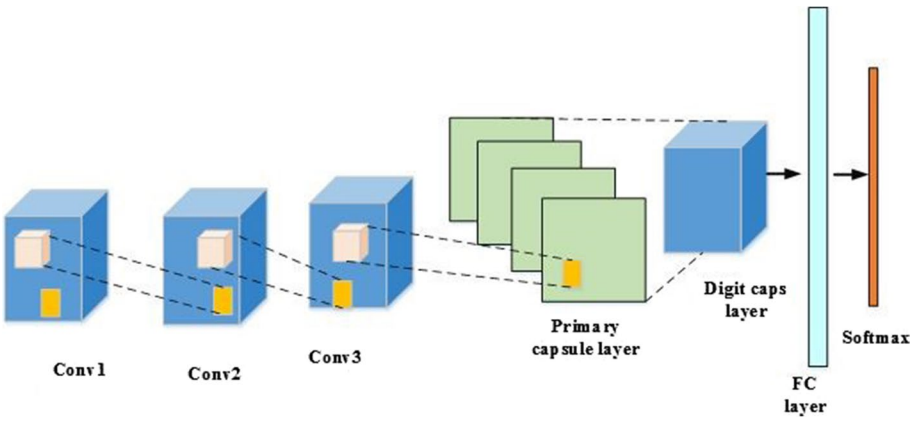


Fig. 4 Architecture of proposed Hybrid DCCAE

Activation function In every convolutional layer, ReLU is applied, and the activation function is used for converting the weighted input sum to an output of nodes. Mathematically, the ReLU is computed as:

$$g(x) = \max(0, x)(ReLU) \tag{28}$$

Dropout layer This layer executes powerful and inexpensive operations that enhance the generalization capability of the network. This approach includes randomly eliminating and restoring neurons on the training with probability declared by the hyper-parameter, known as the dropout rate.

Batch Normalization This layer minimizes the training time and improves the neural network's efficiency. The motive of this layer is to perform a normalization of the input data across the entire input layer.

FC layer FC layer has neurons linked to entire neurons in the prior layer. These layers are utilized as a final element of the classifier, which is sustained using the features extracted by the following convolutional layer. The final layer that generates the network output is a softmax (sigmoid function), and it is expressed as:

$$f(z) = \frac{e^{z_i}}{\sum_k e^{z_k}} \tag{29}$$

The Capsule network has the following layers: convolutional layer, 3 FC layer, Primary Caps layer, and a Digit Caps layer.

Primary caps layer With the help of linear combination, this layer splits the entire feature maps into capsules covering a convolution layer and reshaping operation.

Digitcaps layer This layer outputs 16D (6 and 9) vectors and contains several features. These caps obtain input from 1152 capsules. It involves every instantiation variables need for the object reconstruction.

The input prediction vector of the primary capsule layer is given as:

$$\hat{u}_{k|l} = z_{kl} x_k \tag{30}$$

where z_{kl} is a matrix weight between the capsule k and l in the present and succeeding layer. x_k is capsule output k in the prior layer. The coupling coefficient D_{kl} is represented as:

$$D_{kl} = \frac{e^{B_{kl}}}{\sum_k e^{B_{kl}}} \tag{31}$$

where B_{kl} is a matrix weight among capsule k and l in the present and succeeding layer. The squash function is

$$R_l = \frac{\|R_l\|_2^2 R_l}{\left(1 + \|R_l\|_2\right)\left(1 + \|R_l\|_2\right)} \tag{32}$$

where R_l is an in-between vector in l capsule and $\|R_l\|_2$ is 2- norm of R_l. Finally, the classified images are segmented using Hybrid DCCAE and the forgery image is obtained.

Segmentation using ADFC This clustering easily captures density information and initializes the membership matrix. Using the density for each sample, density peaks declare the number of clusters and automatically initialize the matrix. Let the density of every sample y_j be represented as ρ_j. Then the entire samples are arranged in descending order with densities, and it is represented as $Y_s = \{\overrightarrow{y_j}\}, j = 1, 2 \ldots \ldots m$ and the respective densities are $\{\overrightarrow{\rho_j}\}, j = 1, 2 \ldots \ldots m$. The cut-off density is represented as:

$$s_d = \left(\sum_{j=1}^{m} \rho_j \right) \omega \tag{33}$$

where ω is a rate of density. Then the potential centre of the cluster, whose densities are higher than other samples and it is defined as:

$$P_c = \left\{ \overrightarrow{y_j} \sum_{j=1}^{r-1} \overrightarrow{\rho_j} < s_d, \sum_{j=1}^{r} \overrightarrow{\rho_j} \geq s_d, j = 1, 2 \ldots \ldots r \right\} \tag{34}$$

From Eqs. (32) and (33), it can be seen that the density rate affects the value s_d. Hence, ω is used for adjusting the number of potential clusters. Finally, the suitable initial cluster center is selected from the potential centre of the cluster, and the cluster number c_j is obtained.

In order to avoid that two samples belonging to a similar cluster being chosen as the initial cluster center, there must be a certain distance between different initial cluster centers. The distance between the two initial cluster centres must be higher than a certain threshold. It is represented as:

$$\partial = \left(\frac{1}{\rho} \sum_{j=1}^{r} d_{jk} \right) \alpha \tag{35}$$

where d_{jk} is the distance among x_j and x_k, $\alpha$is a distance rate and adjusted.

The following Equation updates the density function:

$$\rho_i = \sum_{jk} \delta_j \times \varsigma_j \left(d_{jk} - c_d \right) Y \tag{36}$$

where $Y(y) = \begin{cases} 1 & y < 0 \\ 0 & \text{elsewhere} \end{cases}$, c_d is a cut off distance, δ_j is domain density and ς_j is delta distance. The parameters like δ_j and ς_j improves the clustering process. Hence, it is efficiently able to segment the forgery regions. The overall proposed algorithm for blind forgery detection is provided in Table 2.

Hence, by implementing the proposed approach, effective detection accuracy can be obtained in three different datasets. The detection time and feature dimensionality are comparatively low than the previous techniques. So, the proposed research work can be proved to be highly accurate for detecting blind image forgeries effectively. The performance of the proposed system can be analysed clearly in the upcoming section.

Table 2 Algorithm for proposed forgery detection model

Input: Forged Images from CASIA V1 and Grip dataset

Pre-processing using WE-CLAHE

Step 1: Consider the Set of images
Step 2: Represent the original image as M and the estimated image as $M^{'}$
Step 3: Evaluate the error measure using equation (1)
Step 4: Express the minimum error function using equation (2)
Step 5: Histogram computation
Step 6: Evaluate the average number of pixels per region using $Q_A = \frac{Q_x \times Q_y}{Q_g}$
Step 7: Perform histogram clipping
Step 8: Perform clipping limit for histogram height using equation (5)
Step 9: Obtain the entire clipped pixel using $Q_c = (Q_x \times Q_y) - H_t$
Step 10: Construct a cumulative histogram using equation (7)
Step 11: Improve the image quality and brightness using equation (8)
Step 12: \\ Pre-processed image

Feature extraction using H-(DTCWT–DCT) and VGG-16

Step 13: Acquire pre-processed image
Step 14: Represent the first $F_{new}^{(l)}\left(e^{kw}\right)$and second $S_{new}^{(l)}\left(e^{kw}\right)$filter bank response
Step 15: Perform DTCWT to decompose the image $g(a,b)$
Step 16: Obtain 6 CWT functions
Step 17: Divide $g(a,b)$into s non overlapping sub-bands
Step 18: Express DCT transform using equation (11)
Step 19: \\ Extract the coefficients of *H-(DTCWT–DCT)*
Step 20: \\ Extract the features of *VGG-16*

Feature selection using IHH

Step 21: Assign horse movement using equation (12)
Step 22: Express grazing field using equation (13)
Step 23: Express hierarchy law using equation (15)
Step 24: Represent the sociability using equation (17)
Step 25: Represent Imitation, Defence and Roam using equations (19), (21), (23)
Step 26: Represent the velocity of horses using equation (25)
Step 27: Utilize the regulating parameters β_i and γ_i
Step 28: \\ Select meaningful features

Classification by Hybrid DCCAE

Step 29: Initialize CNN network
Step 30: Represent coupling coefficient D_{kl}using equation (31)
Step 31: \\ Classified images

Segmentation using ADFC

Step 32: Denote the cut-off density using $s_d = \left(\sum_{j=1}^{m} \rho_j\right)\omega$
Step 33: update the density function using equation (36)
Step 34: If $y<0$, $Y(y)=1$
Step 35: else
Step 36: $Y(y)=0$
Step 37: End If

Output: Segmented forgery regions

4 Result and discussion

This section presents the results and discussion of forgery detection of the proposed scheme. All implementations were processed on a system with 8 GB RAM and Intel Core i5 CPU with 3.0 GHz speed. The overall implementation is carried out on the Python platform. The robustness of the proposed DCCAE is compared with the recently published research works. The performance of the proposed system in terms of three datasets are analysed in terms of different metrics to evaluate the efficiency of blind image forgery detection. The description of data, performance evaluation metrics and performance analysis is provided in the upcoming sub sections.

4.1 Dataset detail

Three datasets like, GRIP, Coverage, and CASIA V1, are used to evaluate the proposed DCCAE.

GRIP This dataset was introduced by Cozzolino [14], and it comprises 80 plain forgery images and 80 input images. The images are available in 1024x768 in PNG format. The tampered portions are stored separately in PNG format. The location of tampered portions in forgery images are stored in text format. It has three parts: 50 tampered images after translation, 20 sets of an image after scaling and rotation and 50 real images without forgery images. The sample image representation is given in Figure 5.

Coverage This dataset [44] comprises CMFD images and the corresponding real with SGOs (similar but genuine objects). It contains 100 tampered and 100 original images. This dataset is introduced to highlight and overcome tamper detection. In this, pairs of original-forgery images are elucidated with the forged and duplicated segment masks and tampering measures. This dataset works on 6 types of attacks: scaling, translation, rotation, free-for, illumination and combination. The sample image representation is given in Fig. 6.

CASIA V1 This dataset [9] has spliced and CMFD images with an image size of 384x256. It consists of 1721 images (800 original and 921tampered images). The splice images are created using splice regions from one image to another using software called Adobe Photoshop. The sample image representation is given in Fig. 7.

4.2 Performance metrics

To analyze the performance of the developed scheme, some metrics are commonly used. The metrics using the formula are described below.

Accuracy: The overall system's efficiency is determined by the degree of accuracy. For the proposed technique, accuracy is the ratio of accurately identified samples

$$A = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \tag{37}$$



Fig. 5 Sample image representation from the GRIP



Fig. 6 Sample image representation from the Coverage

Precision: The proportion of predicted positive outcomes that are true positives.

$$P = \frac{T_p}{T_p + F_p} \tag{38}$$

Sensitivity: Calculates the exactly determined positive proportions. The formulation of sensitivity measures can be depicted as follows:

$$Se = \frac{T_p}{T_p + F_n} \tag{39}$$

F1-score: The ensemble mean of sensitivity and precision.

$$F1 - score = \frac{2T_p}{2T_p + F_p + F_n} \tag{40}$$

True Positive Rate (TPR): The actual positive probability is tested as positive.

$$TPR = \frac{T_p}{T_p + F_n} \tag{41}$$

True Negative Rate (TNR): The actual negative probability is tested as negative.

$$TNR = \frac{T_n}{T_n + F_p} \tag{42}$$



Fig. 7 Sample image representation from the CASIA V1

False-positive rate (FPR): A proportion of true negatives is misclassified as positives.

$$FPR = \frac{F_p}{T_n + F_p} \tag{43}$$

Where T_p, F_p, F_n and T_n are accurately identified pixels, incorrect pixels predicted correctly, incorrect pixels predicted correctly, and incorrect pixels predicted incorrectly.

4.3 Performance analysis

This section shows the quantitative and qualitative representation of the three datasets. Further, the performance measure of the proposed model is compared with the existing models to test the performance of the proposed model.

Figures 8, 9 and 10 show the visual analysis of the proposed DCCAE, and the images are obtained from the three datasets like Coverage, Grip and CASIA. The representation of input, forgery and GT (ground truth) images are given. Forgery images are compared with ground truth images to verify that the detected forgery region was identified correctly or not. It can be observed that the proposed clustering-based method can correctly detect the forgery regions due to the properties of adaptive density clustering. Table 3 describes the parameter setting of the proposed system.

Setting the parameter values as described in Table 3 shows that the proposed system can gather effective performance in detecting blind image forgeries. By varying the epoch size

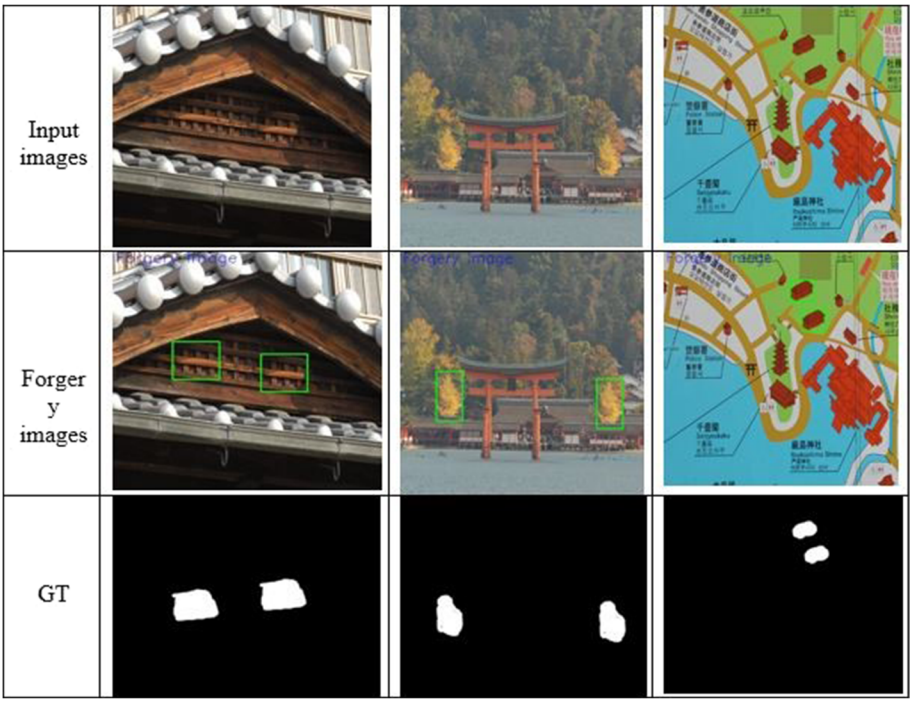


Fig. 8 Quantitative analysis on Grip database

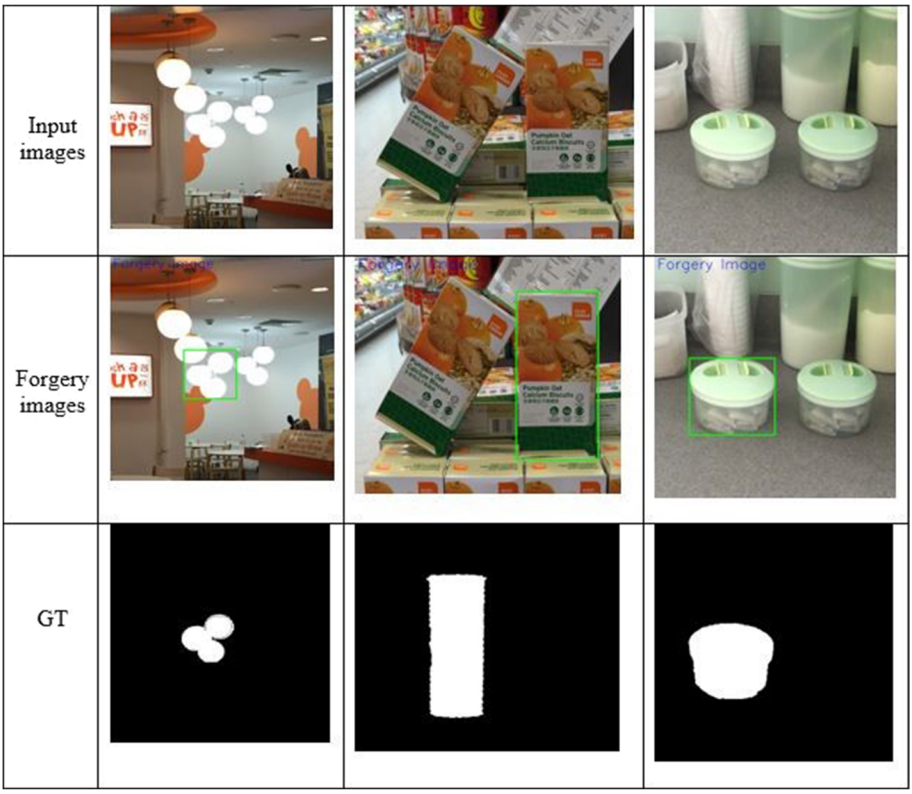


Fig. 9 Quantitative analysis on Coverage database

from 1 to 100, better performance can be attained. The learning rate of 0.01% is set for the rapid model learning and the loss function is used to overcome the cross entropy loss.

Table 4 shows the accuracy and TPR comparison of the approaches like DWT-DRLBR (Discrete Wavelet Transform and dominant rotated local binary patterns) [34], SRM (spatial rich model) [28], Random Sample Consensus (RANSAC) [14], DCT-LBP [38] and the proposed DCCAE. The performance is tested on the CASIA V1 dataset. The accuracy of the proposed DCCAE is 99.23% which is the highest among all the other approaches. Other methods achieved poor results since these methods take more time for training and have high computational complexity. Further, the TPR value of the proposed DCCAE is 99.1%, and the other models obtained lower TPR values.

Table 5 shows the comparison of the various techniques on the Coverage dataset based on the metrics like accuracy, FPR and TPR. The methods like dense filed (DF) method [8], Segmentation CMFD (SCMFD) [23], ZM (Zernike moment) [31], Going deeper CMFD [35], Iterative CMFD [47], HFPM (hierarchical feature point matching) [22] are compared with the proposed DCCAE. The accuracy and TPR value of the proposed model are 98.75% and 98.9%, respectively. The accuracy and TPR value of the other models are less than the proposed model. Further, the FPR value (1.25) is less for the proposed model. It clearly shows that the system has fewer errors. Table 6 indicates the comparison of DL based forgery detection models.

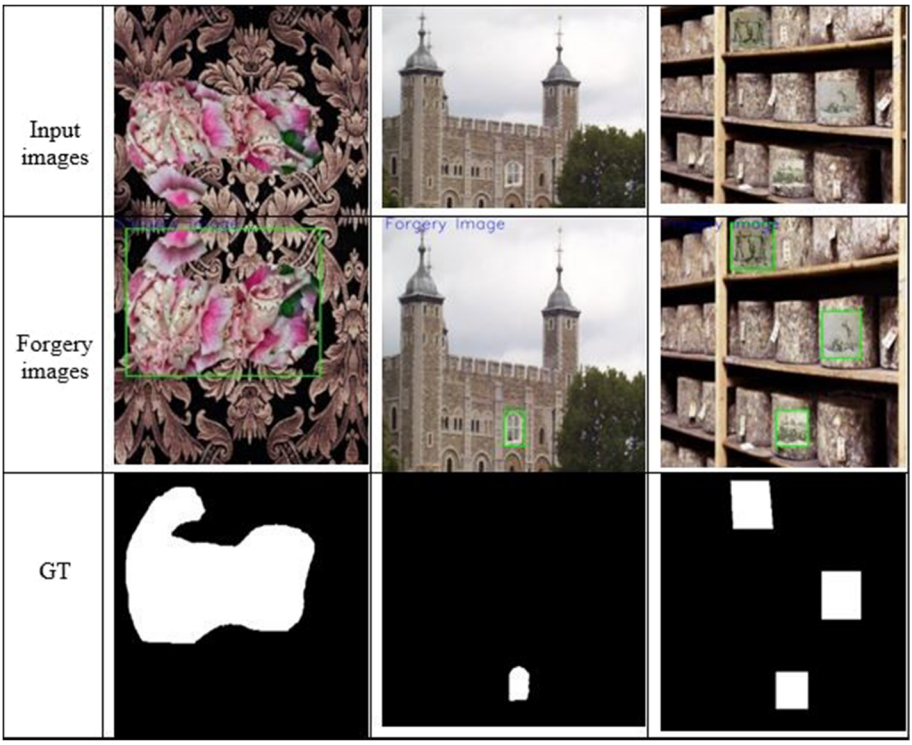


Fig. 10 Quantitative analysis of CASIA dataset

The DL techniques based on blind image forgery detection is compared with the proposed model in terms of accuracy, TPR and FPR. Different models like SVGG Net and Mobile Net v2. A clear observation can be made that the proposed model in terms of three datasets like CASIA VI, Coverage and grip obtains 99.23%, 98.75% and 98.07%, respectively, is far better when compared to the existing DL models. Table 7 indicates the comparison of various approaches to the Coverage dataset.

The performance of the proposed model is compared with different forgery detection techniques like hybrid features and semantic reinforcement network (HFSRNet)

Table 3 Parameter setting for the proposed framework

Sl. No	Parameters	Proposed
1.	Learning rate	0.01%
2.	Epoch Size	100
3.	Loss function	'Binary_ crossen- tropy'
4.	Batch size	128
5.	Number of layers	13
6.	ConV 2D	7
7.	Maxpooling 2D	3
8.	Up sampling 2D	3
9.	Optimizer	Adam

Table 4 Comparison of various techniques on CASIA V1 dataset

Methods	Accuracy (%)	TPR (%)
DWT- DRLBP	98.59	97.87
RANSAC	96.2	93.0
SRM	97.1	97.6
DCT-LBP	97.0	98.24
Proposed	**99.23**	**99.1**

The values highlighted in 'Bold' specify the higher performance outcomes

base-LSTM (Long Short Term memory) Conv, HFSRNet-base-RR (Ringed Residual) Conv, HFSRNet-base-RRConv-SR and HFSRNet-base-LSTMConv-RRConv. When analysing the performance, it can be made obvious that the proposed model, in terms of coverage dataset has obtained a better accuracy rate of 98.75% and AUC performance of 98.2% compared to the existing approaches.

Table 8 displays the GRIP database based detection outcomes. This table demonstrates the performance comparison of the proposed DCCAE with MSFE (Multi-scale feature extraction) [4], Non-overlapping Blocks [36], entropy rate super pixel (ERS) [41] and ASFPM (adaptive segmentation and feature points matching) [27]. The performances like F1 score, precision, and recall are compared. It illustrated that the proposed scheme achieved maximum performance (F1 score of 97.43%, precision of 98.07% and recall of 98.75%). Hence, it is proved that the proposed DCCAE can efficiently detect the forgery with better performance.

ROC curve and the corresponding AUC (Area under the curve) for the three datasets are given in Fig. 11. This curve shows the trade-off between recall and specificity. The near to up left corner represents the best performance than the curve on the right side. It shows that the accuracy of the proposed DCCAE for tampered images is better. The AUC is a metric of the usefulness of the approach, and it is 0.976 for Grip, 0.982 for the coverage dataset and 0.99 for the CASIA V1 dataset, respectively. The ROC plot is designed between True positive rate and false positive rate. The false rate detection is highly superior to the proposed model in case of three datasets. ROC's better performance is attained by the proposed approach, which tends to effective detection of blind image forgeries.

4.4 Statistical test analysis

The effectiveness of the proposed forgery detection model can be estimated by the statistical analysis called Mann-Whitney U test. The Quantitative decision can be offered through statistical analysis and by conducting this, it can be observed that the proposed system shows enhanced performance.

Table 5 Comparison of various techniques on the Coverage dataset

Methods	Accuracy (%)	TPR (%)	FPR (%)
DF	68.68	59.34	21.98
SCMFD	62.09	87.91	63.74
ZM	65.39	46.15	15.38
GDCMFD	59.94	91.21	70.33
ICMFD	52.75	76.92	71.43
HFPM	69.23	80.22	41.76
Proposed	**98.75**	**98.9**	**1.25**

The values highlighted in 'Bold' specify the higher performance outcomes

Table 6 Comparison of DL based forgery detection models [1]

Models	Accuracy (%)	TPR (%)	FPR (%)
SVGG Net	87	87	13
Mobile Net V2	62.09	87.91	63.74
Proposed (CASIA V1)	**99.23**	**99.1**	-
Proposed (Coverage)	**98.75**	**98.9**	**1.25**
Proposed (Grip)	**98.07**%	-	-

The values highlighted in 'Bold' specify the higher performance outcomes

4.5 Mann-Whitney U test

The performance of the proposed model is analysed through a statistical non-parametric test called the Mann-Whitney U test. The work is analysed with and without the utilization of optimization. The significance level of the non-parametric test is 0.05, which is signified as *p*value. If the attained value is lesser than the significance level $p < 0.05$, the technique does not provide better performance. In a similar way, if it is greater than or equal to 0.05, then the technique can attain good performance.

The statistical outcome of the Mann-Whitney U test of the proposed method is attained to be $p = 0.28$. The value p is superior to 0.05, and it proved statistically that the proposed research work achieves better performance. Moreover, the proposed method is established with hypothesis rejection of equal mean.

4.6 Discussion

Recognition of exact blind image forgery is a major requirement in most applications. The blind image forgery detection systems are considered an important approach for improved performance. In this research work, a novel technique for forgery detection is implemented by acquiring data from CASIA VI, Coverage and Grip datasets. The dataset comprises the different images consisting of original and tampered images. For the precise classification and segmentation of forgeries, the proposed model attains 99.23% accuracy for CASIA VI, 98.75% for the coverage dataset, and the GRIP dataset obtained 98.07%. WE-CLAHE performs a pre-processing step for enhancing the image quality, and the most discriminant features are extracted through the adoption of Hybrid DTT and VGGNet. In the selection of optimal features, the IHH approach is used to improve the forgery detection accuracy.

The surveyed methods have shown high time and computational complexity. This can be reduced in the proposed approach by adopting an effective feature selection method to minimize the feature dimensionality. The detection accuracy of forgeries is highly

Table 7 Comparison of various approaches to the Coverage dataset [7]

Models	Accuracy (%)	AUC (%)
HFSRNet-base-LSTMConv	96.8	82.8
HFSRNet-base-RRConv	96.4	89.6
HFSRNet-base-RRConv-SR	97.6	92.5
HFSRNet-base-LSTMConv-RRConv	98.2	94.7
Proposed (Coverage)	**98.75**	**98.2**

The values highlighted in 'Bold' specify the higher performance outcomes

Table 8 Comparison of various approaches to GRIP dataset

Approaches	Precision (%)	Recall (%)	F1 (%)
MSFE [4]	58.2	73.2	64.83
Non-overlapping Blocks [36]	89.30	83.50	87.55
ERS [41]	96.80	92. 26	90. 26
ASFPM [31]	89.195	83.73	86.38
proposed	**98.07**	**98.75**	**97.43**

The values highlighted in 'Bold' specify the higher performance outcomes

improved, and it minimizes the error rate. Due to the accumulation of larger datasets, high storage space is required in the existing works, which can be overcome by reducing the number of features in the proposed method. Due to the degraded performance of learning in the classification, performance outcomes are less, which can be overcome in the proposed approach by adopting an effective learning model.

Different forms of comparative analysis are made by comparing the performance of proposed method with existing approaches in terms of three different datasets. Through the analysis, a better justification can be made that the proposed DL model effectively detects the blind image forgeries with a higher rate of performance outcomes. Compared with the proposed approach, the existing techniques attained fewer performance outcomes. The existing techniques obtain less accuracy rates due to enlarged storage complexities, expansion of error possibilities and slower process due to a greater gathering of datasets.

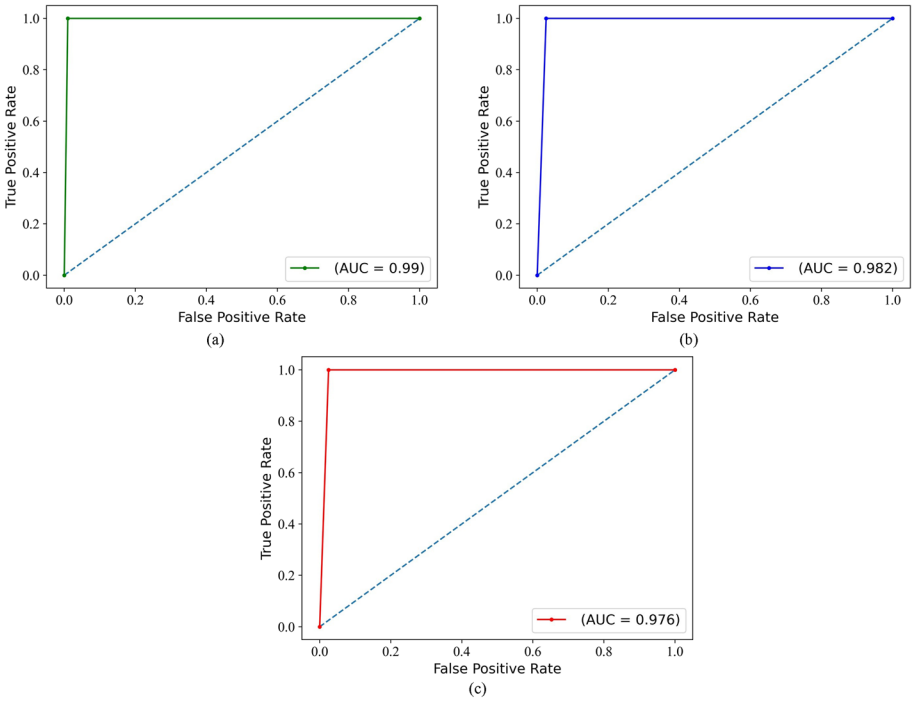


Fig. 11 ROC characteristics (**a**) CASIA V1 (**b**) Coverage (**c**) Grip

5 Conclusion

In recent days, false accusation due to the emergence of fake data creation is rising because of different forgeries. A significant way to overcome the forgeries is by designing of effective detection model. One of the popular image forgery methods is copy move and splicing. In this method, one or many portions are copied and fixed at various positions inside the image. This paper proposes blind forgery detection using various approaches. Initially, the input forgery images are pre-processed using the WE-CLAHE approach for effective noise removal. Significant features are extracted from the clustered data using Hybrid DTT and VGGNet. The feature dimensionality is lessened through IHH to improve the classification accuracy. Finally, the Hybrid DCCAE framework is presented for optimal image forgery recognition. The ADFC approach is presented to segment similar pixel regions. The proposed algorithm proved that this model could efficiently identify the forgery images. The system is trained on original and forgery images. The performance is tested on three benchmark datasets which contain original, forgery and ground truth images. The proposed DCCAE is compared with various approaches based on the measures like accuracy, TPR, precision, FPR, F- score and ROC. The accuracy of the CASIA V1 dataset is gained by 99.23%, the coverage dataset has attained an accuracy of 98.75%, and the GRIP dataset has obtained 98.07% of accuracy correspondingly. The deep learning based proposed approach enhances the performance of forgery detection better than the existing approaches.

The proposed approach has obtained better system performance due to its effective capability and reliability of the training model. Due to the adoption of a better feature extraction model, significant features are gained, whereas the overall dimensionality of the feature is minimized, which adversely improves the training and detection speed. The possibility of error is minimized, and better classification accuracy can be attained.

Although the proposed approach offers a better ability to detect blind image forgeries, it also faces some limitations, such as less convergence rate, over fitting issues, recognition cannot be accurate, and the computation complexity. Hence, the proposed work can be further extended as, the robustness of the proposed DCCAE will be analyzed on various complex databases. Hybrid feature selection methodologies can further improve classification accuracy by selecting the most optimal features and minimising the detection time. Clustering technique can be adopted for accurate localization of forgery regions. The Content based image retrieval system [21] can be employed with the extracted features from the proposed method to attain effective performance. More similar databases can be retrieved accurately through this work which can be done in future.

Authors contributions All authors have equal contributions in this work

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Declarations

Conflict of interest Authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate All the authors involved have agreed to participate in this submitted article.

Consent to publish All the authors involved in this manuscript give full consent for publication of this submitted article.

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